The analytic degree has no positive lower bound in terms of the Cauchy-weight degree

Candidate counterexample packet for arXiv:2208.04006, Section 6.2

11 August 2026

Status: candidate counterexample, likely valid, subject to human review. Near-constant nonconstant entire functions give a full negative answer to the lower-bound question as stated.

1 Source question

The source is Armin Rainer, *Quantitative tame properties of differentiable functions with controlled derivatives*, arXiv:2208.04006v2 and Nonlinear Analysis 237 (2023), 113372 [1].

For $\varepsilon > 0$, the analytic degree $d_f(2\varepsilon)$ of a function holomorphic on $D_{1+2\varepsilon}$ is the least $d \geq 0$ such that

$$\|f\|_I \leq \left(\frac{4|I|}{|E|}\right)^d \|f\|_E \tag{1}$$

for every interval I obtained by intersecting a real affine line in $\mathbb{C}$ with D_1 and every measurable $E \subseteq I$ of positive length.

The source applies Cauchy's estimates and takes

$$M_0 = \frac{\|f\|_{D_{1+\varepsilon}}}{\varepsilon}, \qquad \mu_j = \frac{Cj}{\varepsilon} \quad (j \geq 1), \tag{2}$$

where $C > 0$ is a fixed conversion constant. It proves

$$d_f(2\varepsilon) \leq 10\mathfrak{d}_{2\mu|\infty}(\varepsilon)$$

and then states on printed page 25:

> "We do not know if there is a lower bound for $d_f(2\varepsilon)$ in terms of $\mathfrak{d}_{2\mu|\infty}(\varepsilon)$."

As stated, the answer is no: the weight degree records the Cauchy envelope, but it does not record the distance of f from a nonzero constant.

2 Counterexample

Let

$$j_0(\varepsilon) = \lceil \log \varepsilon^{-1} \rceil_{\mathbb{N}}, \qquad D(\varepsilon) = \mathfrak{d}_{2\mu|\infty}(\varepsilon)$$

for the weight in (2). Directly from the source definition,

$$D(\varepsilon) = \sup\left\{N \in \mathbb{N} : \frac{\varepsilon}{2C} \sum_{j=j_0(\varepsilon)+1}^{N} \frac{1}{j} < e\right\}. \tag{3}$$

In particular,

$$D(\varepsilon) \geq j_0(\varepsilon). \tag{4}$$

Theorem 1. *There is no positive lower bound for $d_f(2\varepsilon)$ depending only on $D(\varepsilon)$, even if f is required to be nonconstant and entire. More precisely, for every fixed $\varepsilon > 0$ with $\varepsilon < 1$ there are nonconstant entire functions f_δ satisfying the Cauchy weight (2) such that*

$$d_{f_\delta}(2\varepsilon) \longrightarrow 0 \quad \text{as } \delta \downarrow 0,$$

while $D(\varepsilon)$ is unchanged. Moreover, these fixed degree values can be chosen arbitrarily large.

Proof. Fix $0 < \varepsilon < 1$ and, for $0 < \delta < 1$, set

$$f_\delta(z) = 1 + \delta z.$$

This function is nonconstant and entire. On the closed unit disk,

$$1 - \delta \leq |f_\delta(z)| \leq 1 + \delta.$$

Consequently, for every admissible I and E in (1),

$$\frac{\|f_\delta\|_I}{\|f_\delta\|_E} \leq \frac{1+\delta}{1-\delta}.$$

Since $E \subseteq I$ implies $4|I|/|E| \geq 4$, the exponent

$$d_\delta = \frac{\log((1+\delta)/(1-\delta))}{\log 4} \tag{5}$$

makes (1) valid for every I, E. Hence

$$0 \leq d_{f_\delta}(2\varepsilon) \leq d_\delta \longrightarrow 0.$$

For each δ, take M_0 and μ_j exactly as in (2). The number M_0 changes with f_δ, but the sequence $\mu_j = Cj/\varepsilon$ does not. The degree $D(\varepsilon)$ depends only on this sequence and on ε, so it is independent of δ. Thus any proposed lower bound

$$d_f(2\varepsilon) \geq L(D(\varepsilon))$$

valid for all such nonconstant functions must satisfy $L(D(\varepsilon)) = 0$.

Finally choose $\varepsilon_n = e^{-n}$. Then $j_0(\varepsilon_n) = n$, so (4) gives $D(\varepsilon_n) \geq n \to \infty$. The preceding fixed-$\varepsilon$ argument forces $L(D(\varepsilon_n)) = 0$ on this unbounded set of degree values. No positive lower bound depending only on the Cauchy-weight degree can exist. $\square$

If constant functions are allowed, the obstruction is even more immediate: $f \equiv 1$ has analytic degree zero for every ε, while the same weight $\mu_j = Cj/\varepsilon$ gives the finite degree (3), which becomes unbounded as $\varepsilon \downarrow 0$. The family $1 + \delta z$ shows that excluding constants does not repair the question.

3 Interpretation, checks, and novelty

The counterexample does not contradict the source's upper bound. That bound uses an envelope of all derivatives and is intentionally uniform over every function controlled by the chosen weight. Such an envelope can be much larger than the actual analytic complexity of a particular function.

A meaningful reverse comparison would need additional normalization, for example a quantitative lower bound on the oscillation of f relative to its supremum, or a minimal function-adapted weight rather than the universal Cauchy weight. No such condition appears in the question as printed.

The bundled checker evaluates (5) along $\varepsilon_n = e^{-n}$ and confirms $D(\varepsilon_n) \geq n$ from the exact definition. The proof itself is elementary and has no computational dependency.

On 11 August 2026, the run registry and all cheap indexes were searched for arXiv:2208.04006 and for the exact analytic-degree lower-bound phrase. Bounded primary-source searches found the arXiv paper, its open-access 2023 publication, and no later answer or normalization of the question. The published version retains the same statement.

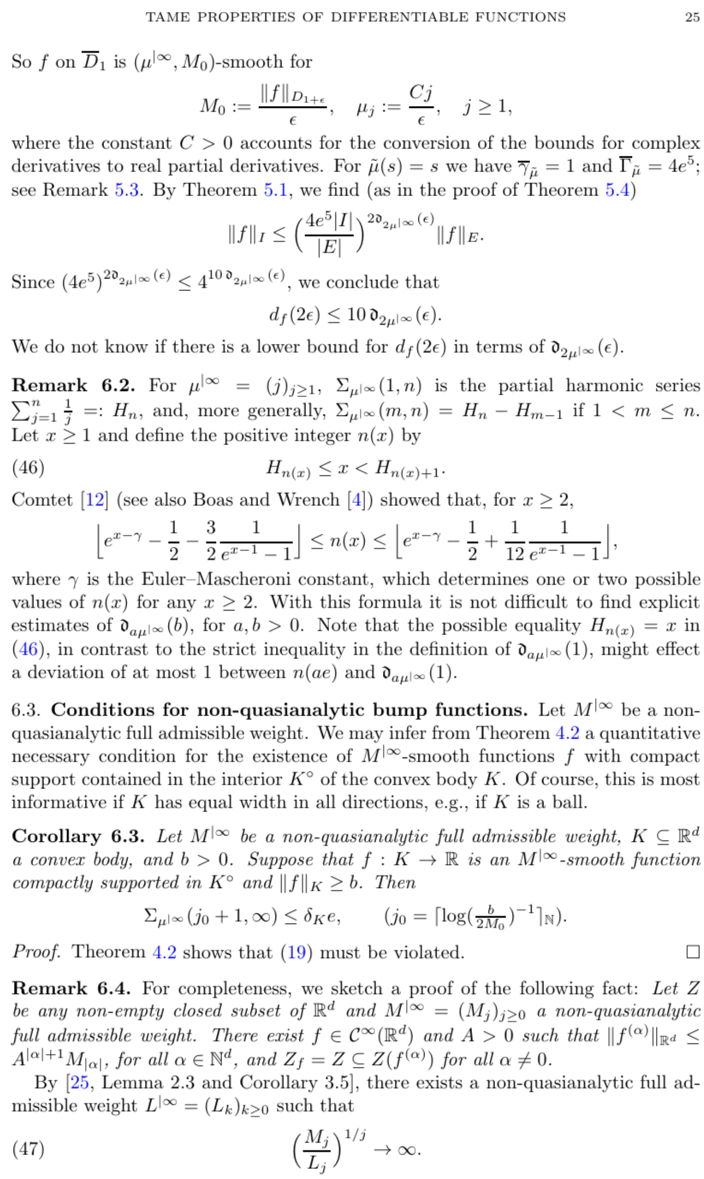
Human review should check that "in terms of" is intended to mean a bound uniform over functions for the Cauchy-derived weight (2). Under that literal and natural reading, the nonconstant counterexample is decisive.

A Source-page evidence

Figure 1 reproduces printed page 25 of the source, including the exact open statement.

References

[1] A. Rainer, *Quantitative tame properties of differentiable functions with controlled derivatives*, arXiv:2208.04006v2, 2023; Nonlinear Analysis 237 (2023), 113372.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So f on $\overline{D}_1$ is $(\mu^{|\infty}, M_0)$-smooth for

$$M_0 := \frac{\|f\|_{D_{1+\epsilon}}}{\epsilon}, \quad \mu_j := \frac{Cj}{\epsilon}, \quad j \ge 1,$$

where the constant $C > 0$ accounts for the conversion of the bounds for complex derivatives to real partial derivatives. For $\tilde{\mu}(s) = s$ we have $\overline{\gamma}_{\tilde{\mu}} = 1$ and $\overline{\Gamma}_{\tilde{\mu}} = 4e^5$; see Remark 5.3. By Theorem 5.1, we find (as in the proof of Theorem 5.4)

$$\|f\|_I \le \Big(\frac{4e^5|I|}{|E|}\Big)^{2\mathfrak{d}_{2\mu^{|\infty}}(\epsilon)} \|f\|_E.$$

Since $(4e^5)^{2\mathfrak{d}_{2\mu^{|\infty}}(\epsilon)} \le 4^{10\,\mathfrak{d}_{2\mu^{|\infty}}(\epsilon)}$, we conclude that

$$d_f(2\epsilon) \le 10\,\mathfrak{d}_{2\mu^{|\infty}}(\epsilon).$$

We do not know if there is a lower bound for $d_f(2\epsilon)$ in terms of $\mathfrak{d}_{2\mu^{|\infty}}(\epsilon)$.

Remark 6.2. For $\mu^{|\infty} = (j)_{j\ge1}$, $\Sigma_{\mu^{|\infty}}(1,n)$ is the partial harmonic series $\sum_{j=1}^n \frac{1}{j} =: H_n$, and, more generally, $\Sigma_{\mu^{|\infty}}(m,n) = H_n - H_{m-1}$ if $1 < m \le n$. Let $x \ge 1$ and define the positive integer $n(x)$ by

$$H_{n(x)} \le x < H_{n(x)+1}. \tag{46}$$

Comtet [12] (see also Boas and Wrench [4]) showed that, for $x \ge 2$,

$$\Big\lfloor e^{x-\gamma} - \frac{1}{2} - \frac{3}{2}\frac{1}{e^{x-1}-1}\Big\rfloor \le n(x) \le \Big\lfloor e^{x-\gamma} - \frac{1}{2} + \frac{1}{12}\frac{1}{e^{x-1}-1}\Big\rfloor,$$

where γ is the Euler–Mascheroni constant, which determines one or two possible values of $n(x)$ for any $x \ge 2$. With this formula it is not difficult to find explicit estimates of $\mathfrak{d}_{a\mu^{|\infty}}(b)$, for $a, b > 0$. Note that the possible equality $H_{n(x)} = x$ in (46), in contrast to the strict inequality in the definition of $\mathfrak{d}_{a\mu^{|\infty}}(1)$, might effect a deviation of at most 1 between $n(ae)$ and $\mathfrak{d}_{a\mu^{|\infty}}(1)$.

6.3. **Conditions for non-quasianalytic bump functions.** Let $M^{|\infty}$ be a non-quasianalytic full admissible weight. We may infer from Theorem 4.2 a quantitative necessary condition for the existence of $M^{|\infty}$-smooth functions f with compact support contained in the interior K° of the convex body K. Of course, this is most informative if K has equal width in all directions, e.g., if K is a ball.

Corollary 6.3. *Let $M^{|\infty}$ be a non-quasianalytic full admissible weight, $K \subseteq \mathbb{R}^d$ a convex body, and $b > 0$. Suppose that $f : K \to \mathbb{R}$ is an $M^{|\infty}$-smooth function compactly supported in K° and $\|f\|_K \ge b$. Then*

$$\Sigma_{\mu^{|\infty}}(j_0+1, \infty) \le \delta_K e, \qquad (j_0 = \lceil \log(\tfrac{b}{2M_0})^{-1} \rceil_{\mathbb{N}}).$$

Proof. Theorem 4.2 shows that (19) must be violated. □

Remark 6.4. For completeness, we sketch a proof of the following fact: *Let Z be any non-empty closed subset of $\mathbb{R}^d$ and $M^{|\infty} = (M_j)_{j\ge0}$ a non-quasianalytic full admissible weight. There exist $f \in \mathcal{C}^\infty(\mathbb{R}^d)$ and $A > 0$ such that $\|f^{(\alpha)}\|_{\mathbb{R}^d} \le A^{|\alpha|+1} M_{|\alpha|}$, for all $\alpha \in \mathbb{N}^d$, and $Z_f = Z \subseteq Z(f^{(\alpha)})$ for all $\alpha \ne 0$.*

By [25, Lemma 2.3 and Corollary 3.5], there exists a non-quasianalytic full admissible weight $L^{|\infty} = (L_k)_{k\ge0}$ such that

$$\Big(\frac{M_j}{L_j}\Big)^{1/j} \to \infty. \tag{47}$$

Figure 1: Source paper, printed page 25: the analytic-degree comparison and open lower-bound question.